Chapter

RELATIONS

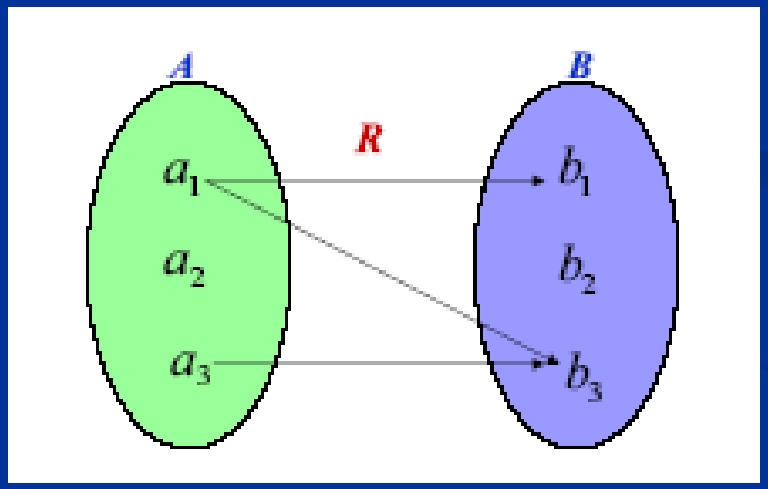
5. Relations

5.1 Relations and Their Properties
5.2 Representing Relations
5.3 Equivalent Relations. Congruences.
Arithmetic Operations on $\mathbf{Z}_n$
5.4 Partial Orderings. Hasse Diagram

5.1.1 Definitions

Definition. A ***binary relation*** from a set A to a set B is a subset of the Cartesian (Cross) Product: $R \subseteq A \times B$.

Notation: $a\ R\ b$



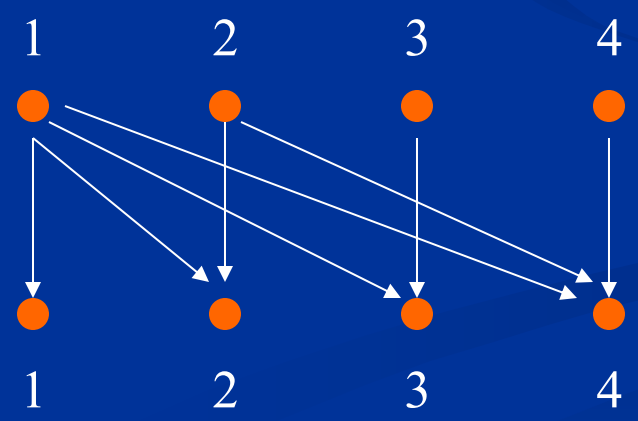
$$R = \{ (a_1, b_1), (a_1, b_3), (a_3, b_3) \}$$

Example. Let $A = \{1, 2, 3, 4\}$, and

$$R = \{(a, b) \in A \times A \mid a \text{ divides } b\}$$

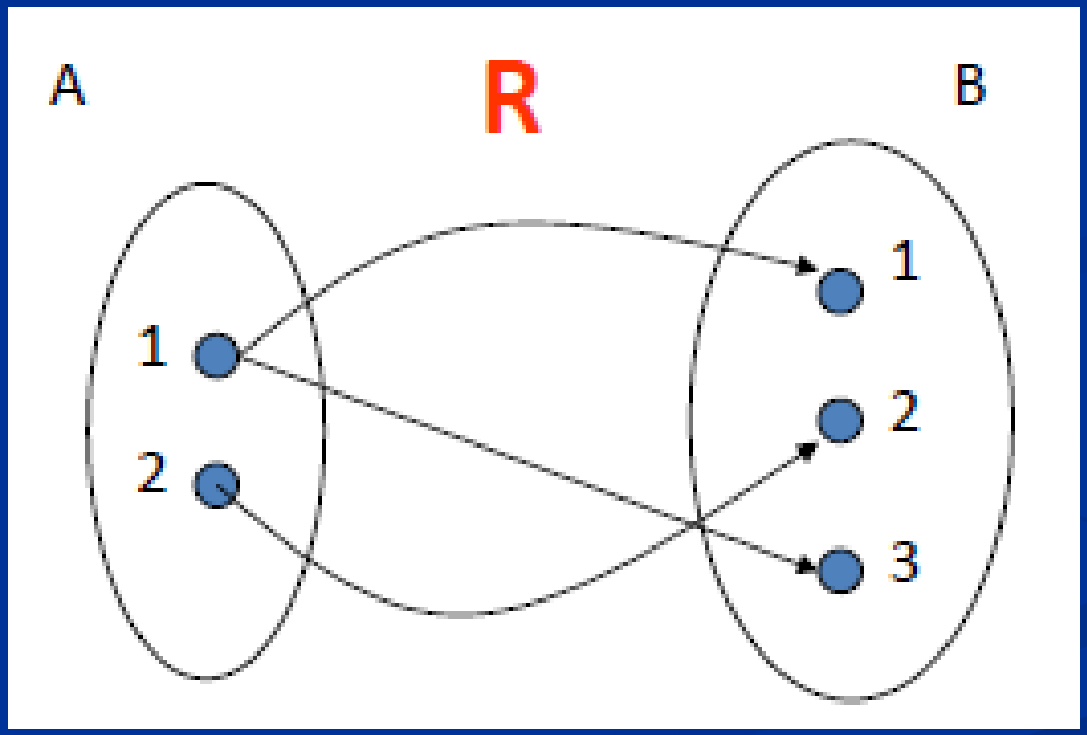
Then R consists of the pairs:

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4,4)\}$$



- **Example**: Let A = {1; 2} and B = {1; 2; 3}, and the relation is given by

$$(x; y) \in R \Leftrightarrow x - y \text{ is even.}$$



2. Properties of Relations

Let R be a relation on A

1) Reflexive : $(a, a) \in R$ for all $a \in A$

2) Symmetric : $\forall\, a, b \in A\,,\quad (a\,R\,b) \rightarrow (b\,R\,a)$

3) Antisymmetric :

$$\forall\, a, b \in A,\ (a\,R\,b) \wedge (b\,R\,a) \rightarrow (a = b)$$

4) Transitive :

$$\forall a,b,c \in A,\quad (a\,R\,b) \wedge (b\,R\,c) \rightarrow (a\,R\,c)$$

Example.

- The relation $\leq$ on **Z** is not symmetric.
- However it is antisymmetric, reflexive.

3. Representing Matrix of Relation

$A = \{a_1, a_2, \ldots, a_m\}$, $B = \{b_1, b_2, \ldots, b_n\}$,

The ***representing matrix*** of $R \subset A \times B$:

$$\mathbf{M}_R = [m_{ij}] \qquad m \times n \text{ matrix}$$

$$m_{ij} = \begin{cases} 0 & \text{if } (a_i, b_j) \notin R \\ 1 & \text{if } (a_i, b_j) \in R \end{cases}$$

R be a relation on a set A, then the matrix $\mathbf{M}_R$ that represents R is a ***square matrix***

Example. Let R be the relation from $A = \{1, 2, 3\}$ to $B = \{1, 2\}$ such that $a\ R\ b$ if $a > b$.

Then the representing matrix of R is

	1	2
1	0	0
2	1	0
3	1	1

- R is ***reflexive*** if and only if all ***diagonal entries*** of $\mathbf{M}_R$ are equal to 1: $m_{ii} = 1$ for all i

	u	v	w
u	1	1	0
v	0	1	1
w	0	0	1

- R is ***antisymmetric*** if and only if $\mathbf{M}_R$ satisfies:

$$m_{ij} = 0 \text{ or } m_{ji} = 0 \quad \text{if } i \neq j$$

	u	v	w
u	1	0	1
v	0	0	0
w	0	1	1

- Let R be a relation on a set A, then the matrix $\mathbf{M}_R$ that represents R is a ***square matrix***
- R is symmetric if and only if $\mathbf{M}_R$ is ***symmetric***

$$m_{ij} = m_{ji} \qquad \text{for all } i, j$$

	u	v	w
u	1	0	1
v	0	0	1
w	1	1	0

Composition of Relations

Given R in $A \times B$, and S in $B \times C$, the composition of R and S is a relation on $A \times C$ defined by

$$R \circ S = \{(a, c) \in A \times C \mid \exists\, b \in B, aRb \text{ and } bSc\}.$$

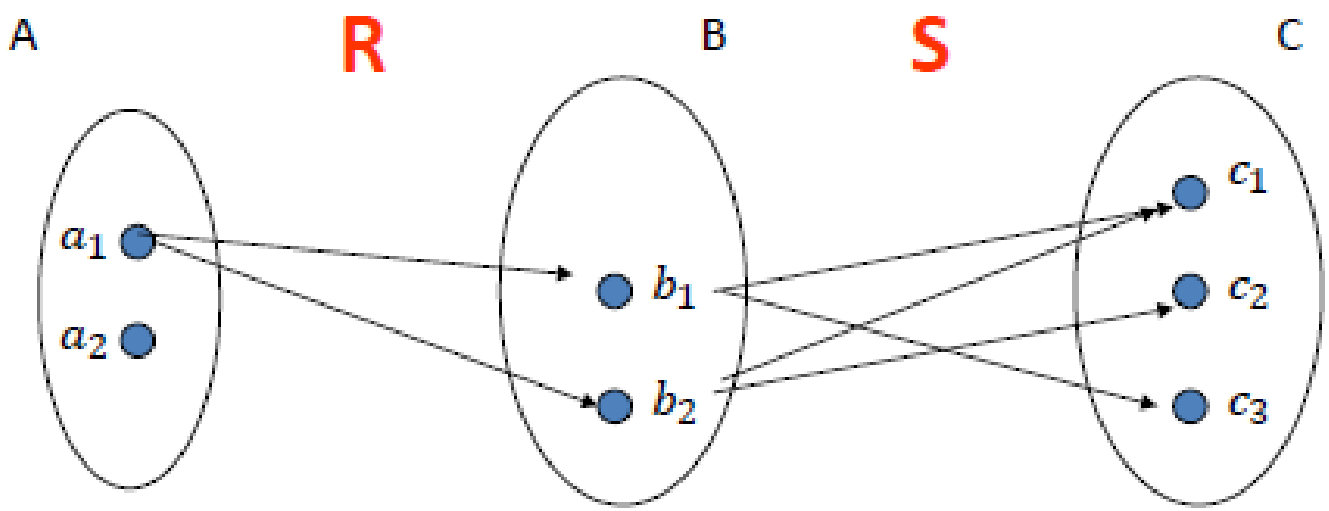
Example. $A = \{a_1, a_2\}, B = \{b_1, b_2\}, C = \{c_1, c_2, c_3\}$

- $R = \{(a_1, b_1), (a_1, b_2)\}$
- $S = \{(b_1, c_1), (b_2, c_1), (b_1, c_3), (b_2, c_2)\}$
- What is $R \circ S$?

- $R \circ S = \{(a_1, c_1), (a_1, c_3), (a_1, c_2)\}$

Graphically

- **Example.** $A = \{a_1, a_2\}, B = \{b_1, b_2\}, C = \{c_1, c_2, c_3\}$
- $R = \{(a_1, b_1), (a_1, b_2)\}$
- $S = \{(b_1, c_1), (b_2, c_1), (b_1, c_3), (b_2, c_2)\}$
- $R \circ S = \{(a_1, c_1), (a_1, c_3), (a_1, c_2)\}$



Definition. Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ zero-one matrix and $\mathbf{B} = [b_{jk}]$ be an $n \times p$ zero-one matrix.

Then the ***Boolean product*** $\mathbf{A} \odot \mathbf{B} = [c_{ik}]$ is the $m \times p$ zero-one matrix defined by

$$c_{ik} = (a_{i1} \wedge b_{1k}) \vee (a_{i2} \wedge b_{2k}) \vee \ldots \vee (a_{in} \wedge b_{nk})$$

$$\text{for } 1 \leq i \leq m \text{ , } 1 \leq k \leq p$$

Note. The $(i, k)^{\text{th}}$ entry of $\mathbf{A} \odot \mathbf{B}$ is obtained by:

- ✓ carrying out the AND operator to every element of the ith row of the matrix $\mathbf{A}$ to the corresponding element of the kth column of $\mathbf{B}$,
- ✓ then take the OR operator on n obtained results

Theorem. The representing matrix $\mathbf{M}_{S \circ R}$ of the relation $S \circ R$ is precisely the matrix $\mathbf{M}_R \odot \mathbf{M}_S = [t_{ik}]$, where

$$t_{ik} = (r_{i1} \wedge s_{1k}) \vee (r_{i2} \wedge s_{2k}) \vee \ldots \vee (r_{in} \wedge s_{nk})$$

$$\text{for } 1 \leq i \leq m \text{ , } 1 \leq k \leq n$$

Example:

Find the matrix representing the relations $S \circ R$, where the matrices representing R and S are

$$\mathbf{M}_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Solution: The matrix for $S \circ R$ is

$$\mathbf{M}_{S \circ R} = \mathbf{M}_R \odot \mathbf{M}_S = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}.$$

Equivalence Relation

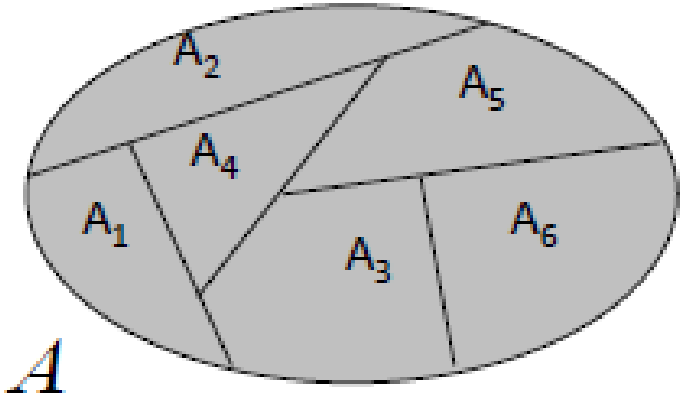
A relation R on a set A is an equivalence relation if

1. R is reflexive: $\forall x \in A, xRx$
2. R is symmetric: $\forall x \, \forall y \; xRy \rightarrow yRx$
3. R is transitive: $\forall x \, \forall y \, \forall z \; xRy \wedge yRz \rightarrow xRz$

Equivalence class of a in A: [a] ={x ∈ A | aRx}
for R an equivalence relation.

Equivalence Classes

Partition of a set A:



$$A_i \cap A_j = \phi \quad \text{whenever} \quad i \neq j$$

$$A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6 = A$$

Equivalence classes of A form a partition of A.

Integers mod n (I)

$$a \equiv b \pmod{n} \leftrightarrow a=qn+b$$

≡ (mod n) is an equivalence relation:

1. ≡ (mod n) is reflexive: $\forall x \in A, x \equiv x \bmod n$
2. ≡ (mod n) is symmetric: $\forall x\ \forall y\ x \equiv y \pmod{n} \rightarrow y \equiv x \pmod{n}$
3. ≡ (mod n) is transitive: $\forall x\ \forall y\ \forall z\ x \equiv y \pmod{n} \wedge y \equiv z \pmod{n} \rightarrow x \equiv z \pmod{n}$.

Partial order

Definition. A relation R on a set A is a ***partial order*** if it is ***reflexive***, ***antisymmetric*** and ***transitive***.

We often denote a partial order by $\prec$

The pair $(A, \prec)$ is called a ***partially ordered set*** or a ***poset***

Reflexive: $a \prec a$

Antisymmetric: $(a \prec b) \wedge (b \prec a) \rightarrow (a = b)$

Transitive: $(a \prec b) \wedge (b \prec c) \rightarrow (a \prec c)$

Definition. A relation R on a set A is a ***partial order*** if it is reflexive, antisymmetric and transitive.

Example. The divisibility relation " | "on the set of positive integers is a partial ordering, i.e. $(\mathbf{Z}^+, |\,)$ is a poset

Reflexive? Yes, $x \mid x$ since $x = 1 \cdot x$

Transitive? Yes?

$a \mid b$ means $b = ka$, $b \mid c$ means $c = jb$.
Then $c = j(ka) = jka$: $a \mid c$

Example. The divisibility relation " | "on the set of positive integers is a partial ordering, i.e. ($\mathbf{Z}^+$, |) is a poset

Antisymmetric?

Yes?

$a \mid b$ means $b = ka$, $b \mid a$ means $a = jb$.
Then $a = jka$
It follows that $j = k = 1$, i.e. $a = b$

Example. Is ($\mathbf{Z}$, |) a poset?

Not a poset.

Antisymmetric?

No

3|-3, and -3|3,

but $3 \neq -3$.

Ex. Is $(2^S, \subseteq)$, where 2^S the set of all subsets of S, a poset?

Yes, A poset.

Reflexive?

Yes, $A \subseteq A$, $\forall A \in 2^S$

Transitive?

$A \subseteq B$, $B \subseteq C$. Does that mean $A \subseteq C$?

Yes

Antisymmetric?

$A \subseteq B$, $B \subseteq A$. Does that mean $A = B$?

Yes

Definition. Let $(S, \prec)$ be a poset. Let a and b are elements of S. If $a \prec b$ then we say that b is an ***upper bound*** of a and a is a ***lower bound*** of b.

The elements a and b are said to be ***comparable*** if either a is a lower bound of b or b is a lower bound of a. Otherwise, they are said to be ***incomparable***.

A poset $(S, \prec)$ such that every two elements are comparable is called a ***totally ordered set***

We also say that $\prec$ is a ***total order*** or a ***linear order*** on S

Example. The relation "$\leq$" on the set of positive integers is a total order.

The elements a and b are ***comparable*** if either a is a lower bound of b or b is a lower bound of a.

Otherwise, they are ***incomparable***.

A poset $(S, \prec)$ such that every two elements are comparable is called a ***totally ordered set***

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Example. The divisibility relation " | "on the set of positive integers is not a total order,

since the elements 5 and 7 are not comparable

Hasse Diagrams

A poset can be represented visually using a special kind of graphs called the ***Hasse diagram***

To define the Hasse diagram we need the concept of direct upper bound.

Definition. An element b in a poset $(S, \prec)$ is said to be a ***direct upper bound*** of an element a in S if

b is an upper bound of a, and there is no upper bound c of a such that

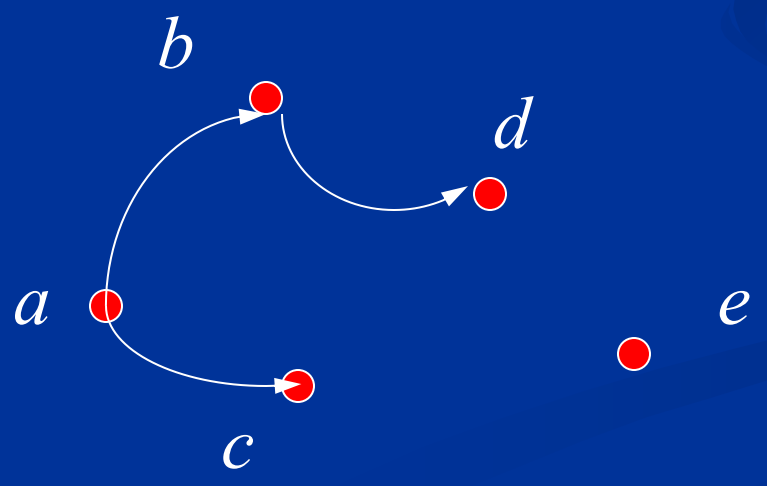
$$a \prec c \prec b, \quad a \neq c \neq b$$

Hasse Diagrams

- Now the ***Hasse diagram*** of a finite poset $(S, \prec)$ is the graph:

✓ whose vertices are points in the plane in one-to-one correspondence with S,

✓ two vertices a, b are joined by an arc directed from a to b if b is a direct upper bound of a



$a \prec b \prec d, \quad a \prec c$

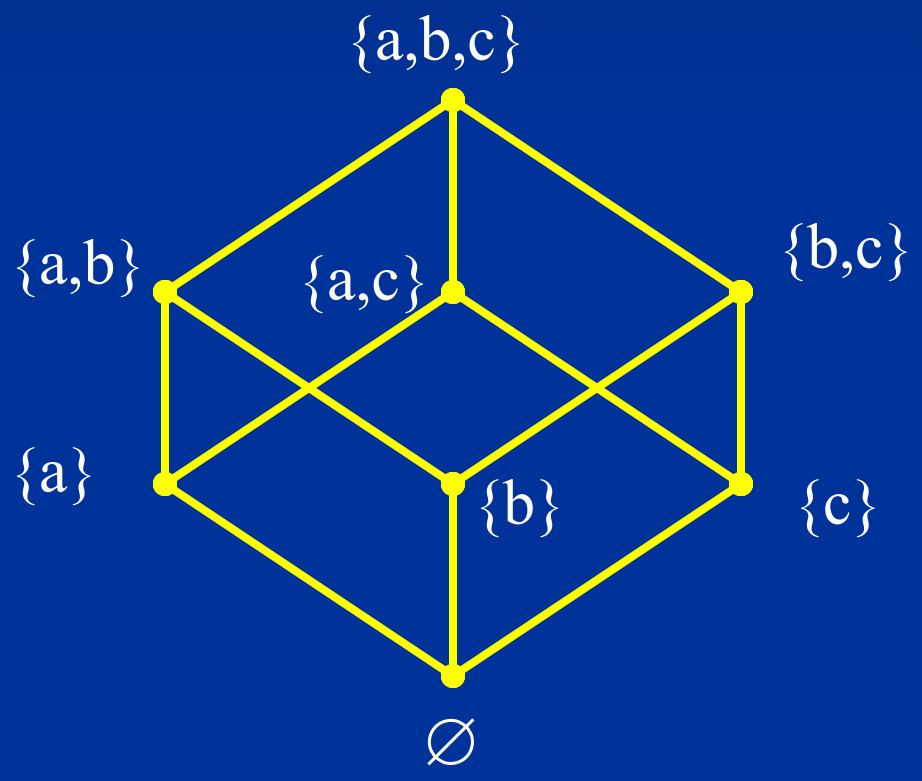
Hasse Diagrams

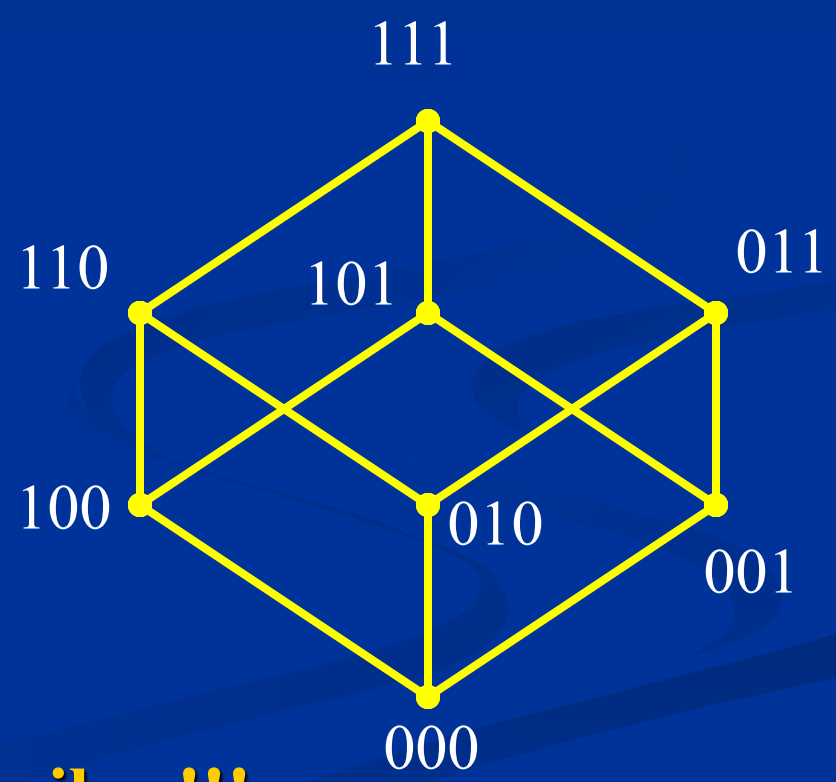
Ex. The Hasse diagram of the poset $(\{1,2,3,4\}, \leq)$ can be drawn as

Note. We did not draw up arrows for the arcs by adopting the convention that arcs are always directed upward

Example. The Hasse diagram of P({a,b,c})

and the Hasse diagram of the set of bit strings of length 3 with natural bitwise order



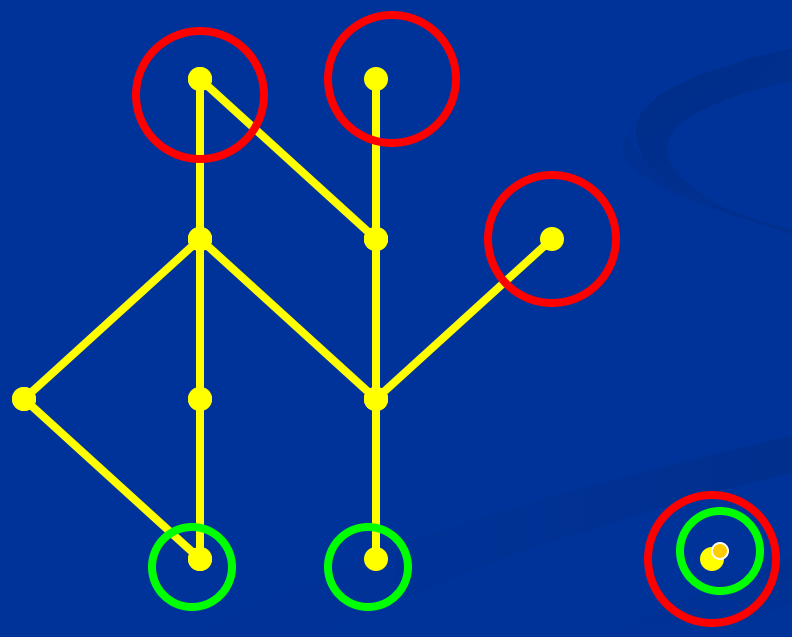


They look similar !!!

Maximal & Minimal Elements

Consider this poset:

- ✓ Each Red is ***maximal***: there is no proper upper bound
- ✓ Each Green is ***minimal***: there is no proper lower bound
- ✓ There is no arc starting from a maximal element
- ✓ There is no arc ending at a minimal element

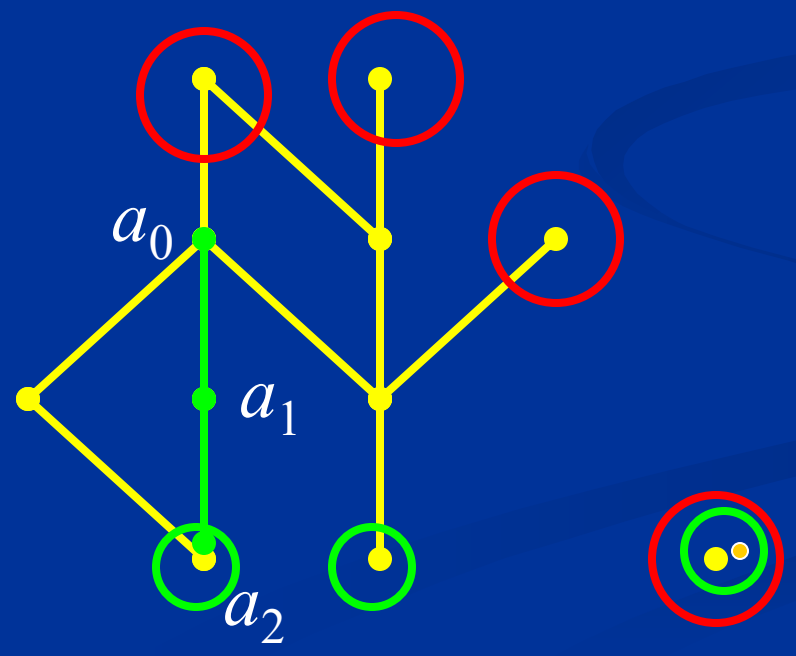


Note. In a finite poset S, maximal and minimal elements always exist.

- ✓ In fact, we can start from any element $a_0 \in S$.

  If a_0 was not minimal, then there exists $a_1 \prec a_0$, and so on until a minimal element is found.

- ✓ The maximal elements are found in a similar way.

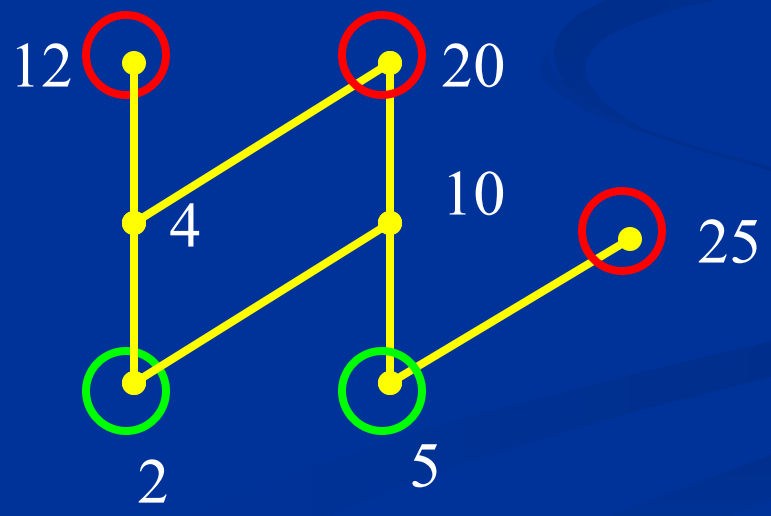


Example. What are the maximal and minimal elements of the poset ({2, 4, 5, 10, 12, 20, 25}, |) ?

Solution. From the Hasse diagram, we see that 12, 20, 25 are maximal elements

and 2, 5 are minimal elements

Thus the maximal and minimal elements of a poset are not necessarily unique



In fact we have

Theorem. In a finite poset, if the maximal element is unique, then it is the greatest element .
Similarly for the least element.

Proof. Let g be the unique maximal element.

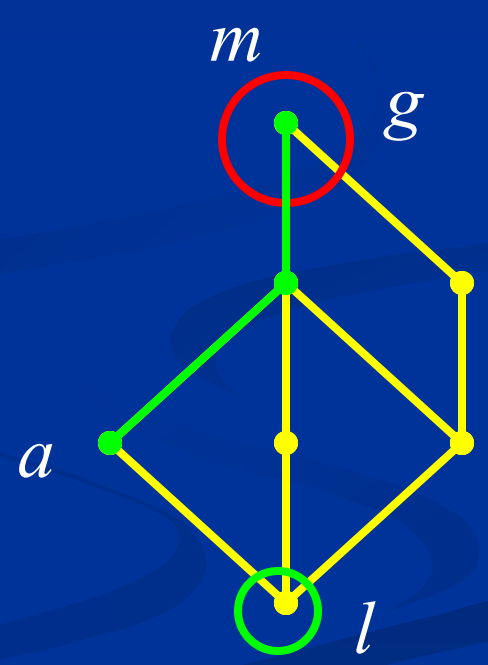
Let a be an arbitrary element, then there is a maximal element m such that

$$a \prec m$$

Since g is unique we must have $m = g$, i.e. $a \prec g$

Therefore g is the greatest element.

Similar proof for the existence of the least element l



Exercise 76. Consider the relation R on $\mathbb{Z}$, given by $aRb \iff a - b$ divisible by n. Is it symmetric?

Exercise 77. Consider a relation R on any set A. Show that R symmetric if and only if $R = R^{-1}$.

Exercise 78. Consider the set $A = \{a, b, c, d\}$ and the relation

$$R = \{(a, a), (a, b), (a, d), (b, a), (b, b), (c, c), (d, a), (d, d)\}.$$

Is this relation reflexive? symmetric? transitive?

Exercise 79. Consider the set $A = \{0, 1, 2\}$ and the relation $R = \{(0, 2), (1, 2), (2, 0)\}$. Is R antisymmetric?

Exercise 80. Are symmetry and antisymmetry mutually exclusive?

Exercise 81. Consider the relation R given by divisibility on positive integers, that is $xRy \leftrightarrow x|y$. Is this relation reflexive? symmetric? antisymmetric? transitive? What if the relation R is now defined over non-zero integers instead?

Exercise 82. Consider the set $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$. Show that the relation $xRy \leftrightarrow 2|(x - y)$ is an equivalence relation.

Exercise 83. Show that given a set A and an equivalence relation R on A, then the equivalence classes of R partition A.

Exercise 84. Consider the set $A = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and the relation

$$xRy \leftrightarrow \exists c \in \mathbb{Z},\ y = cx.$$

Is R an equivalence relation? is R a partial order?